

# SVM & Clustering

INFO 521 · Module 6 · Lecture B — k-means & the Unsupervised  
Turn

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

# Learning outcomes

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By the end of Lecture B you should be able to:

1. **Pose the clustering problem** — structure in  $\mathbf{X}$ , no  $\mathbf{y}$ .
2. **Run k-means** — assign / update — and state its distortion objective  $\mathcal{L}$ .
3. **Diagnose its failure modes** — local optima, choosing  $k$ , scaling.
4. **Refuse the free lunch** — clusters are *not* labels; say why with data.

# The unsupervised turn

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Every method so far consumed pairs  $(\mathbf{x}_n, y_n)$ . Delete the labels:

$$\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathbb{R}^D \quad \text{— no } y_n, \text{ no "right answer".}$$

- **Clustering** — do the patients form *groups*? (today)
- **Dimensionality reduction** — do the features hide a *simpler description*? (Module 7)

Without  $\mathbf{y}$  there is no test-set truth to check against — evaluation becomes *internal* (fit to structure) plus *judgment* (is it meaningful?).

# k-means: the objective

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Choose  $K$  **centroids**  $\mu_k$  and an assignment  $z_n \in \{1..K\}$  for each patient, minimizing the **distortion** — a loss again, so  $\mathcal{L}$ :

$$\mathcal{L}(\{z_n\}, \{\mu_k\}) = \sum_{n=1}^N \|\mathbf{x}_n - \mu_{z_n}\|_2^2$$

Two interlocking sub-problems, each trivial with the other fixed:

- **Assign** (fix  $\mu$ ): send each  $\mathbf{x}_n$  to its nearest centroid.
- **Update** (fix  $z$ ): move each centroid to the **mean** of its members — the squared-error-minimizing point (m1a's loss, yet again).

Alternate until nothing moves. Each step lowers  $\mathcal{L} \Rightarrow$  **guaranteed to converge** — though not necessarily to the best answer.

# k-means on NHANES: the adiposity plane

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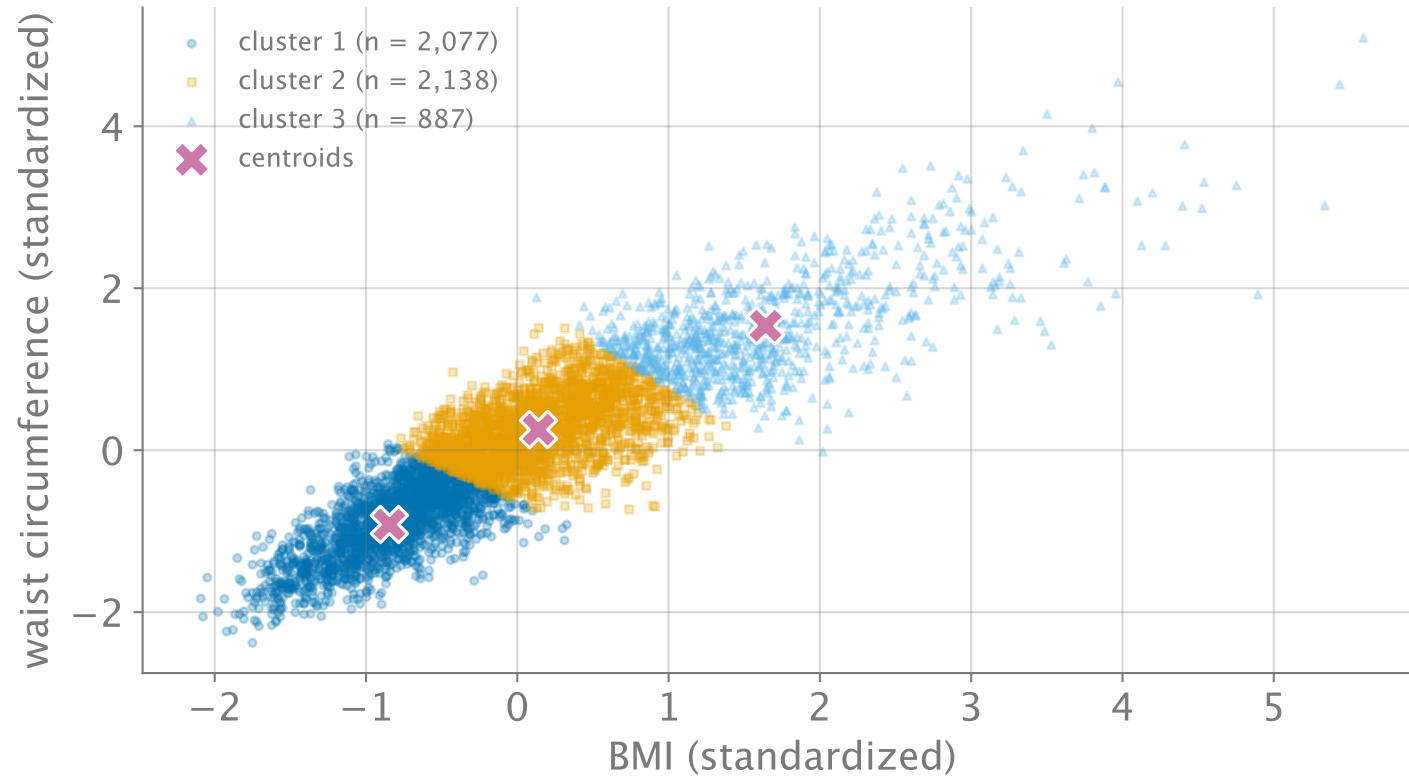


Figure 1: k-means ( $K = 3$ , from scratch, seed 521) on standardized (BMI, waist): the algorithm slices the adiposity continuum into bands; purple X = centroids.

# Standardize first — distances have units

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k-means lives on **Euclidean distance**, so the feature scales *are* the model:

- Raw units: waist (cm, SD  $\approx 17$ ) would out-shout HbA1c (% , SD  $\approx 1$ ) hundreds-fold — clustering would be “waist, with commentary.”
- Standardizing (each feature to mean 0, SD 1) gives features an equal vote — the same move as m1a’s standardized age, now **load-bearing**.
- No distance-based method is scale-free: choosing units **is** choosing the answer.

# Local optima: restarts are not optional

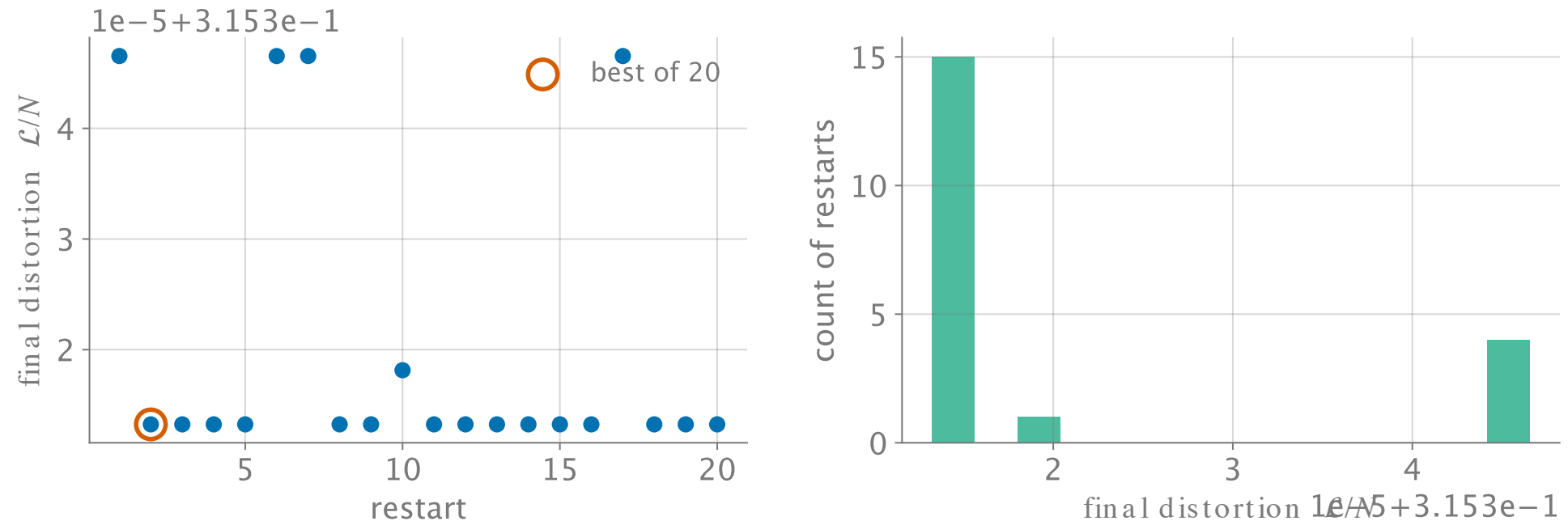


Figure 2: 20 random initializations,  $K = 4$ , same data: k-means lands on different local optima — keep the restart with the lowest  $\mathcal{L}$ .

# Choosing K: the elbow, honestly

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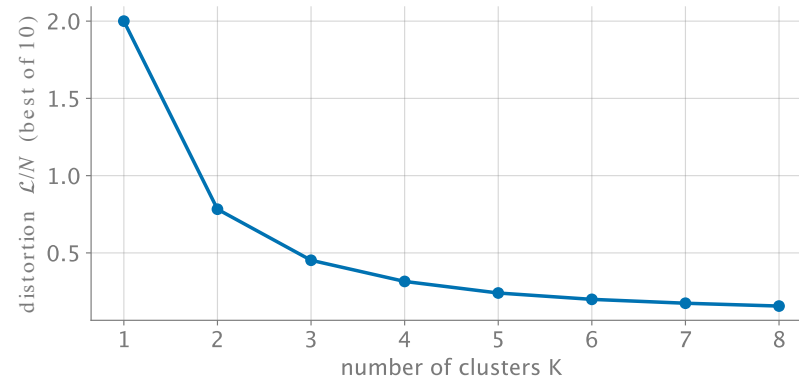


Figure 3: Distortion vs.  $K$  (best of 10 restarts each): always decreasing — the question is where it stops being worth it. Here: a soft elbow, not a verdict.

- $\mathcal{L}$  **always** falls with  $K$  (at  $K=N$  it is zero) — m2a's training-error trap, wearing clusters.
- Look for the **elbow** where marginal gain collapses... and accept that real data often gives a *soft* one.  $K$  is a modeling **choice**, owned and argued.



# Hard assignments → soft: the GMM bridge

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k-means gives every patient exactly one cluster — even on the fence. The probabilistic upgrade: a **Gaussian mixture model**,

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

fit by **EM** — the same assign/update dance, made soft:

- **E-step**: *responsibilities*  $r_{nk} = p(z_n=k \mid \mathbf{x}_n)$  replace hard  $z_n$  (Bayes' rule, m4a, inside the loop).
- **M-step**: weighted MLE updates (m3a) for  $\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ .
- k-means **is** the  $\boldsymbol{\Sigma}_k \rightarrow \varepsilon \mathbf{I}$  limit — hard assignment is a shrunk-covariance special case.

# Clusters are not diagnoses

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Color the K = 3 adiposity clusters by the **held-back** hypertension label:

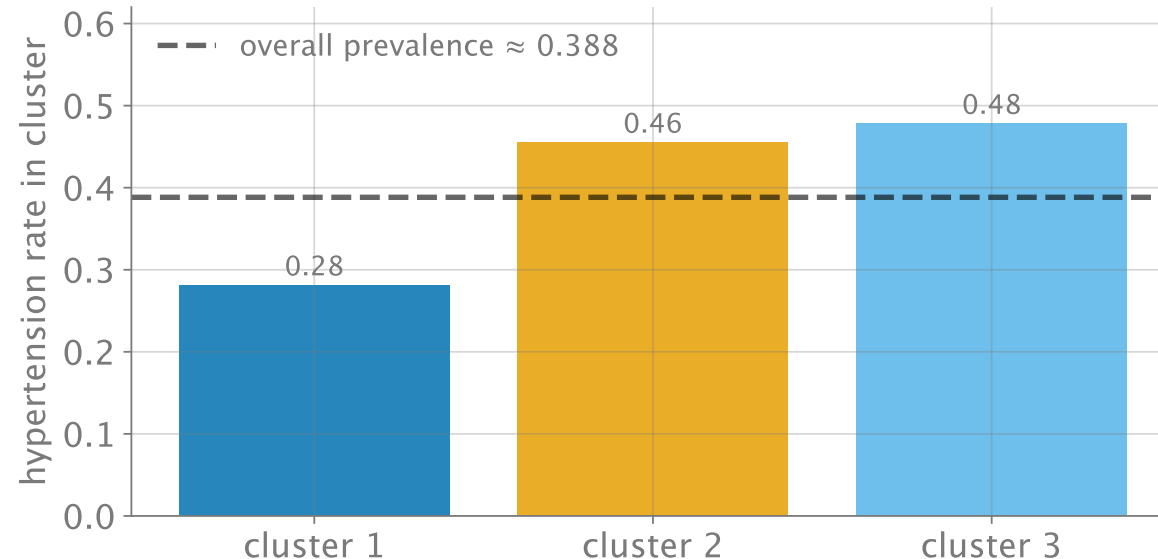


Figure 4: Hypertension rate inside each K = 3 cluster vs. overall prevalence (dashed): adiposity clusters ENRICH risk — none of them IS the diagnosis.

- The clusters **never saw** blood pressure (leakage rule, m1a) — yet risk climbs across them: unsupervised structure is *related* to the outcome...
- ...but no cluster is remotely pure. **Pretending clusters are labels manufactures diagnoses the data never made.** Project 2's honesty requirement, in one figure.

# Live demo: the k-means explorer

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Step assign / update by hand in the week05 tool — trap it in a local optimum, then rescue it with a restart.

# Explorer — static fallback

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*The iframe above does not print or export to PDF; this snapshot stands in.*

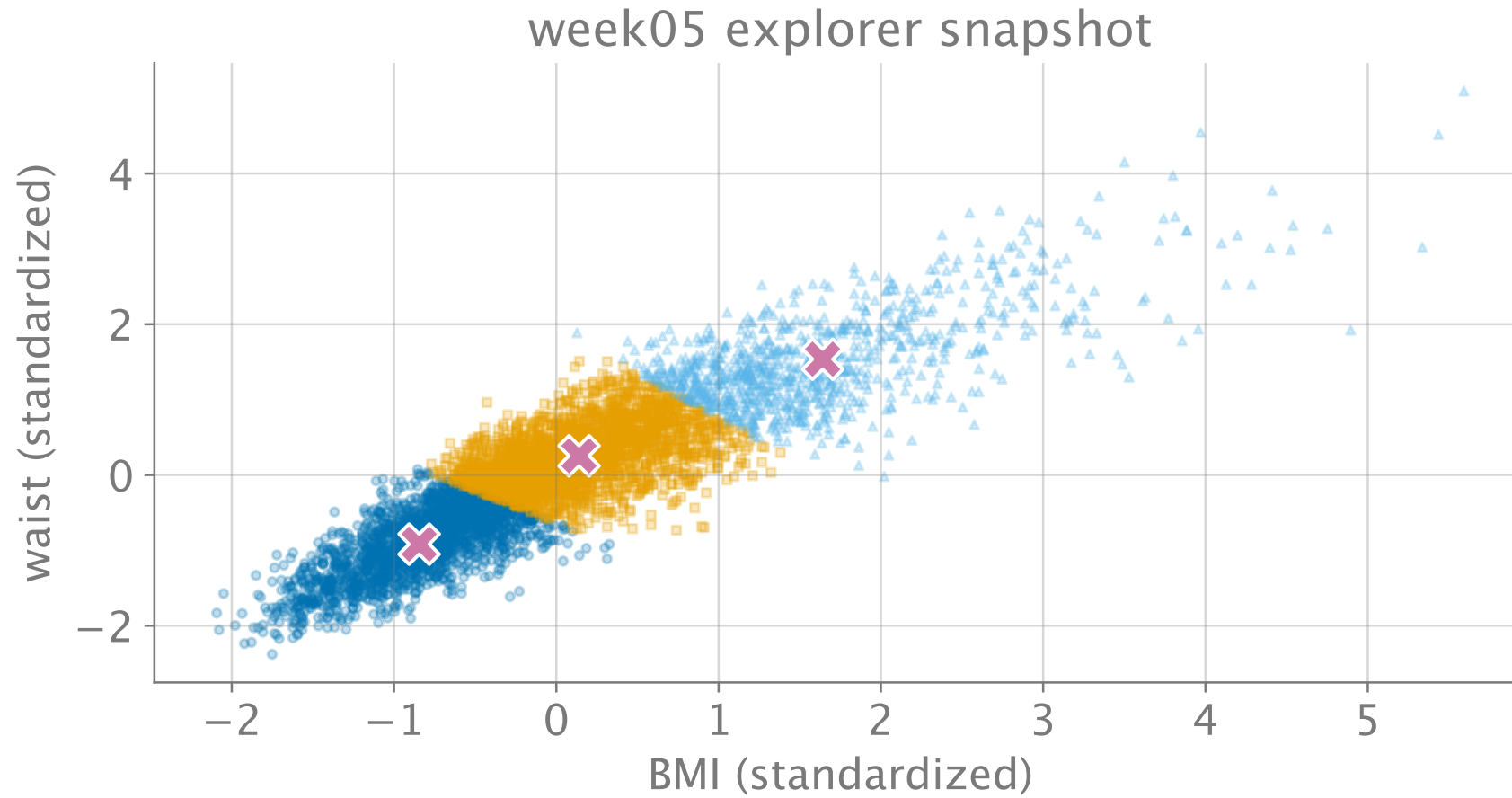


Figure 5: Static fallback: k-means on the NHANES adiposity plane (used in print / PDF).

# Recap & bridge

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- **k-means**: assign / update on distortion  $\mathcal{L}$ ; converges, but to a local optimum — restart and keep the best.
- **$K$  and scaling are choices** — the elbow suggests; you decide and defend.
- **GMM/EM**: the soft, probabilistic upgrade; k-means is its shrunk-covariance limit.
- **Clusters  $\neq$  labels**: enrichment (0.30  $\rightarrow$  0.48) without purity — describe, don't diagnose.

**Next (Module 7):** the other unsupervised question — not “which groups?” but “which *directions* matter?” PCA, and the course's closing synthesis.